



Armada Kalamunda Group

Early Pregnancy Loss Guideline

Table of Contents

1. Purpose	2
2. Applicability	2
3. Background.....	2
4. Guideline	2
4.1 Assessment	2
4.2 Confirmation of Pregnancy	2
4.3 History.....	3
4.4 Physical Examination.....	3
4.5 Investigation.....	4
4.6 Correlation of Ultrasound with Serum β -hCG Measurements	5
4.7 Haemodynamically Unstable Woman	5
4.8 Ectopic Pregnancy	6
4.8.1 Risk Factors for Ectopic Pregnancy	6
4.8.2 Clinical Presentation.....	7
4.8.3 Diagnosis of Ectopic Pregnancy.....	8
4.8.4 Management of Ectopic Pregnancy.....	8
4.9 Women with Non-Viable Intrauterine Pregnancy (IUP).....	8
4.9.1 Expectant Management	9
4.9.2 Medical Management	9
4.9.3 Misoprostol Protocol (outpatient regime for early pregnancy loss < 14 weeks).....	10
4.9.4 Surgical Management	10
4.10 Rh D Immunoglobulin for Sensitising Event Immunoprophylaxis.....	10
4.11 Early Pregnancy Assessment Service (EPAS)	11
4.12 Early Pregnancy Loss Support Resources	12
5. Legislative context.....	12
6. Supporting information	13
7. Compliance, monitoring and evaluation.....	Error! Bookmark not defined.

8. Relevant standards	13
9. References	13
10. Document owner	14
11. Review	14
12. Authorisation	14

1. Purpose

The purpose of this guideline is to clinically diagnose the woman's presenting complaint in a timely manner, initiate treatment, and make appropriate referrals for further care where necessary.

2. Applicability

This guideline applies to women in the first trimester of pregnancy – up to 13 weeks of completed pregnancy.

This guideline applies to all clinical staff working in the Emergency Department within the Armadale Kalamunda Group (AKG).

3. Background

Miscarriage and ectopic pregnancy can cause significant maternal morbidity and mortality. Miscarriage occurs in at least 15-20% of pregnancies¹; 1-2% of reported pregnancies are ectopic².

Significant psychological effects can occur in women and their families who suffer miscarriage; however appropriate [support](#) during and after the event can have positive lasting effects³.

4. Guideline

4.1 Assessment

Assessment and diagnosis can be made through a combination of patient history, physical examination and clinical investigation. Refer to [Flowchart - Assessment of Suspected Early Pregnancy Loss](#).

Initial assessment of haemodynamic stability is essential. If haemodynamically unstable, refer to the [Haemodynamically Unstable Woman](#) section of this guideline.

Ultrasound scan (USS) should be performed as soon as possible, and urgently if clinically indicated³.

4.2 Confirmation of Pregnancy

All women of reproductive age presenting with a history of recent/current abdominal pain, pelvic pain, shoulder tip pain and/or per vaginum (PV) bleeding, syncope or signs of shock should have an urgent serum pregnancy test performed (irrespective of LMP, contraception, history of sterilization or reported sexual inactivity)^{3,4,5}.

Always consider the possibility of [ectopic pregnancy](#) in a woman with abdominal pain and/or vaginal bleeding with a positive pregnancy test^{2,3}

4.3 History

History should consider^{4,5}:

- Pain assessment: site, onset, character, radiation, associated symptoms (particularly vomiting, syncope, shoulder tip pain), timing, exacerbating/relieving factors, severity
- Vaginal bleeding:
 - Onset and any precipitants e.g. sexual activity
 - Nature and severity e.g. spotting, heavy bleeding, clots, frequency of change of menstrual product
 - Passage of tissue or products of conception (POC)
 - Last Menstrual Period (LMP) duration and nature, menstrual history
- Reproductive history:
 - Contraception use and method
 - History of recent assisted conception (method, fresh/frozen cycle)
 - If known pregnancy, what investigations already performed (especially USS, sight report if possible)
- Obstetric history:
 - Gravida - total number of pregnancies regardless of outcome
 - Parity - number of previous pregnancies carried over 20 weeks gestation
 - Outcome (miscarriage, ectopic, termination, stillbirth, live birth), and gestation carried to, for each pregnancy
 - Pregnancy complications, delivery methods
- Gynaecological history:
 - Previous Pelvic Inflammatory Disease or Sexually Transmitted Infections
 - Previous gynaecological surgery e.g. Dilatation and curettage, tubal surgery/ligation
 - Previous infertility
 - Cervical Screening Test/Pap smear history
 - [Risk factors for ectopic pregnancy](#)

4.4 Physical Examination

Physical examination should include^{4,5}:

- Base line observations (PR, BP, RR, Temperature)

If patient presents as haemodynamically unstable refer to the [Haemodynamically Unstable Woman](#) section of this guideline

- Abdominal examination to check for:
 - Tenderness (Rigidity & Guarding)
 - Distention
 - Any surgical scars
- Vaginal examination (individualize as clinically indicated):

- Speculum examination:
 - Source & amount of bleeding
 - Evidence of POC in cervical canal (remove & send for histopathology)
- Bimanual examination:
 - Cervical excitation (motion tenderness)
 - Open or closed cervical os (can only be assessed by digital examination, not speculum)⁴
 - Assess for adnexal masses (ectopic pregnancy or other masses)
 - Size of uterus related to menstrual dates

4.5 Investigation

Bedside tests:

- Send serum β -hCG (quantitative pregnancy test) rather than performing a Urine β -hCG
- Urine β -hCG (qualitative pregnancy test) can give a false negative result if β -hCG level below detection threshold
- High sensitivity urine test detection threshold is 20-50 IU/L depending on brand; this can miss ectopic pregnancies with β -hCG below the test threshold

Laboratory:

- Send blood for serum β -hCG, FBC and Blood Group & Antibody screen
- Consider Venous Blood Gas (VBG), U+E, X-match, Coagulation studies +/- ROTEM if significant haemorrhage
- Consider from history whether requires screening for blood borne and infectious diseases⁴ e.g. Hep B/C, HIV, Chlamydia/Gonorrhoea PCR (1st pass urine or swab), Syphilis serology

Ultrasound:

- Transvaginal scanning (TVS) by an experienced sonographer is the gold standard and should be utilised wherever possible^{2,3,4,5}
- If TVS unavailable or declined by patient, transabdominal ultrasound (TAS) may be used; recognising that it is less accurate, with poorer resolution (particularly at lower gestational age) compared with TVS for diagnosis of early pregnancy complications³
- Utilise point of care ultrasound (**PoCUS**) at the bedside in an emergency e.g. evaluating for ectopic pregnancy in a haemodynamically unstable patient – is there a definite intra-uterine pregnancy (IUP), signs of free fluid
 - Use of PoCUS in ED to influence clinical decision should be within the limits of clinician's accreditation e.g. eFAST or early pregnancy ultrasound
- If patient has already had an ultrasound in current pregnancy confirming IUP then no urgent need to perform repeat scan in ED (unless haemodynamically unstable or other significant pathologies need to be excluded e.g. appendicitis, ovarian torsion)
 - Request outpatient USS (via PRC or GP)
 - USS to be performed (and results available) prior to EPAS appointment
- Consider the psychological distress that a delay to ultrasound and EPAS appointment may cause to the patient; provide with [patient resources](#) as appropriate

4.6 Correlation of Ultrasound with Serum β -hCG Measurements

Refer to [Table – Correlation of Ultrasound with Serum \$\beta\$ -hCG Measurements](#)

USS assessment & serial serum β -hCG enables the clinician to ascertain the location and potential viability of an early pregnancy⁵.

- A single β -hCG indicates when an IUP should be visualised on ultrasound⁴
- A single β -hCG value does not differentiate between a viable and nonviable pregnancy
- Serial β -hCG is useful in diagnosis of asymptomatic ectopic pregnancy, or to assess viability of a pregnancy

For a potentially viable IUP up to 6-7 weeks gestation the following applies:

- Mean doubling time for β -hCG is 1.4-2.1 days⁶
- 85% of pregnancies show serial β -hCG rise of at least 66% every 48 hours⁷
- Minimal expected increase over 48 hours is 53%⁷

The earliest definitive evidence of pregnancy visible on ultrasonography is the gestational sac.

- In 8-31% of women, it may not be possible to confirm if pregnancy is intrauterine or extrauterine at first visit²
- 4 weeks + 3 days is the earliest time to see gestational sac within the endometrium on TVS⁴

The **discriminatory zone** is the serum β -hCG level above which a gestational sac should be visible on ultrasound if an IUP is present (usually visible on TVS when gestational sac ≥ 3 mm)⁵.

This corresponds to a discriminatory zone for β -hCG of:

- Approximately 1500 IU/L on TVS
- Approximately 6500 IU/L on TAS⁴

The absence of an intrauterine gestational sac at β -hCG concentrations above the discriminatory zone strongly suggests an ectopic pregnancy⁸.

Caution required in^{4,5}:

- Multiple gestations: There is no proven discriminatory zone for β -hCG
- Presence of fibroids: USS may be less reliable
- Assisted conception: Higher risk of multiple gestations, ectopic and heterotopic pregnancies
 - Heterotopic pregnancy: co-existing IUP with ectopic pregnancy
 - Natural conception rate: 1/30,000 pregnancies
 - Assisted reproduction rate: 1/100 pregnancies⁹
- Diagnosis of complete miscarriage: ensure adequate follow-up in case of undiagnosed ectopic

4.7 Haemodynamically Unstable Woman

Haemodynamically unstable women require urgent intervention; presume ruptured ectopic pregnancy or incomplete miscarriage with cervical shock or massive haemorrhage⁵.

- Resuscitate as required considering airway, breathing and circulation
- Call for help as appropriate e.g. ED Consultant, Duty Anaesthetist

- Ensure sufficient IV access (e.g. 2 x 16-gauge IV cannulae)
- Perform urgent speculum examination and remove POC as clinically indicated. This may stop bleeding and restore blood pressure
- Urgent bedside ultrasound (PoCUS) to assess for presence of free fluid e.g. eFAST
- Obtain urgent gynaecological review concurrently with resuscitation
 - Urgent surgical review required if underlying cause not established
- Consider Tranexamic Acid (1g IV over 10 mins) and activation of [Massive Transfusion Protocol](#)
- If bleeding persists and ectopic has been excluded, consider:
 - Ergometrine (250 micrograms slow IV injection or 500 micrograms IM) **OR**
 - Oxytocin (5-10 units slow IV injection or 10 units IM) if history of heart disease, thyrotoxicosis or severe vascular disease **OR**
 - Misoprostol 800-1000 micrograms sublingual or buccal
- If decision made to take unstable patient to Operating Theatre (OT):
 - Inform Nurse Manager to organise OT team
 - Continue resuscitation en route to OT
 - Transfer to theatre should not be delayed by attempts to establish a normal circulating plasma volume
- A haemodynamically unstable patient is a clinical indication for:
 - Surgical evacuation of the uterus for incomplete miscarriage
 - Laparoscopy and / or laparotomy for removal of ectopic pregnancy

4.8 Ectopic Pregnancy

Refer to [Flowchart - Management of Ectopic Pregnancy](#)

Failure to promptly diagnose and manage an ectopic pregnancy can be catastrophic

- Important cause of maternal morbidity and mortality worldwide:
 - 80% of first trimester maternal deaths¹⁰
 - 10-15% of all pregnancy related maternal deaths¹¹
- Commonly (95%) occurs in the fallopian tube; other less common locations include caesarean/hysterotomy scar, cornual, ovarian, cervical, abdominal and heterotopic pregnancies¹⁰
- Falling or stationary β -hCG does not exclude the risk of rupture following surgical, medical or expectant management
- Risk of rupture exists until β -hCG fallen to $\leq 5\text{IU/L}$ ⁴

4.8.1 Risk Factors for Ectopic Pregnancy

Some factors are associated with increased likelihood of ectopic pregnancy, however a third of women with ectopic pregnancy will have no known risk factors^{3,5}:

- Previous ectopic pregnancy (incidence recurrent ectopic ~15%; 30% after two)
- Previous tubal surgery

- Infertility (risk increases with length of)
- Assisted reproductive technology
- Intrauterine contraceptive device in situ
- Previous pelvic/genital infection
- Documented tubal damage/pathology
- Cigarette smoking
- Age 40 years or older
- Sterilisation (risk increased nine-fold)

4.8.2 Clinical Presentation

Ectopic pregnancy clinical manifestations can occur anytime post ovulation and sexual intercourse. Clinical manifestations typically appear six to eight weeks after the last normal menstrual period, but can occur earlier, or later, especially if the pregnancy is not in the fallopian tube⁴.

Common symptoms³:

- Amenorrhoea (not universal)⁸
- Vaginal bleeding (75% of women)⁸ – can be irregular vaginal bleeding (spotting)
- Abdominal or pelvic pain (80-90% women)

Other symptoms^{2,3}:

- Normal pregnancy discomforts may be present e.g. breast tenderness, nausea, frequent urination
- Shoulder tip or diaphragmatic pain in 10-20% of women with ruptured ectopic
- Dizziness/syncope
- Urinary or gastrointestinal symptoms e.g. diarrhoea or pain on defecation
- 9% may have no symptoms before rupture

More common clinical signs^{2,3}:

- Lower abdominal/pelvic/adnexal tenderness or mass
- A third of women – no clinical signs⁴
- Physical examination is often unremarkable in a woman with a small, un-ruptured ectopic pregnancy⁴

Other reported clinical signs^{2,3}:

- Cervical motion tenderness⁶
- Rebound tenderness or peritoneal signs
- Pallor
- Abdominal distension
- Tachycardia and hypotension as a result of profound intraperitoneal haemorrhage
- Reflex bradycardia caused by vagal response to intraperitoneal blood or cervical shock
- Shock/collapse
- Orthostatic hypotension

4.8.3 Diagnosis of Ectopic Pregnancy

The most appropriate diagnostic criteria include a combination of⁸:

- Positive pregnancy test
- Empty intrauterine cavity
- Complex adnexal mass +/- extrauterine gestational sac

Ultrasound should be performed by an experienced ultrasonographer. Suspected ectopic pregnancy is an indication for calling in the on-call ultrasonographer out of hours.

In a haemodynamically unstable patient utilise [PoCUS](#). Do not delay urgent gynaecology referral and resuscitation to wait for departmental ultrasound.

If patient is unstable with free fluid in abdomen/pelvis need to consider possible ectopic pregnancy even if IUP seen ([heterotopic pregnancy](#)).

If suspected ruptured ectopic on USS or unstable refer to [Haemodynamically Unstable Woman](#)

4.8.4 Management of Ectopic Pregnancy

Urgent referral to Gynaecology team at Armadale Hospital for:

- Decision on further management and disposition
- Expectant, medical and surgical management options
- Choice will depend on the clinical situation and patient preference

4.9 Women with Non-Viable Intrauterine Pregnancy (IUP)

Miscarriages should be classified as follows¹³:

- **Complete miscarriage** – Previously confirmed IUP where all tissue from the pregnancy has been passed; Ultrasound Scan (USS) shows no apparent products; bleeding generally settles
- **Incomplete miscarriage** - A confirmed non-viable pregnancy on ultrasound with bleeding where some products of conception (POC) have passed but some POC remain in the uterus
- **Missed miscarriage** - A confirmed non-viable pregnancy on ultrasound with intact gestation sac. The cervix is closed and no products of conception have been passed

Expectant, medical and surgical management options are available for a woman with a non-viable IUP. Refer to [Flowchart - Management of Non-Viable Intrauterine Pregnancy](#).

Women should be given an explanation of each option, what to expect and provided with written information/patient information leaflets.

Explanation may be given by either ED or Gynaecology dependent on what is most appropriate for the situation.

Other factors to take into account when explaining options¹³:

- Most first trimester miscarriages occur completely and spontaneously without any intervention
- Incomplete miscarriage (and inevitable miscarriage where products of conception are completely in utero with a dilated cervix) - high success rates occur with expectant (75-94%) and medical management (96-100%)

- Missed miscarriage - expectant management is more likely to fail (14-50% success) compared with medical management (80-90% success)
- Women treated expectantly have more outpatient visits and follow-up than those treated medically, and those treated medically have more visits than surgically treated women
- Women treated medically have more bleeding but less pain than those treated surgically
- The method of miscarriage management does not affect subsequent fertility

[Anti-D immunoglobulin](#) should be given (if indicated) in Rhesus negative women.

If discharged home women should be informed of indications to return to hospital:

- Pain unrelieved by simple analgesia
- Syncope
- Soaking of one sanitary pad within 60 minutes (should be advised to avoid tampons)

Follow up care should be with the GP for contraception and general advice

4.9.1 Expectant Management

Expectant management should be offered to women who prefer this option as there is reduced risk of pelvic infection; however, some women may still require medical or surgical management. For women with an intact sac, resolution may take several weeks, and overall efficacy rates are lower.

- Woman should have easy accessibility to hospital Emergency Department
- Advise that medical/surgical management can still be chosen at a later date if desired
- eReferral to the [Early Pregnancy Assessment Service \(EPAS\)](#) at Armadale Hospital (including those with a complete miscarriage, unless β -hCG ≤ 5 IU/L)
- Provide with safety netting and [discharge information](#)

4.9.2 Medical Management

Misoprostol is the preferred prostaglandin in the medical management of miscarriage in the first trimester. Vaginal Misoprostol is more effective than oral Misoprostol, but vaginal route should be avoided if bleeding and/or signs of pelvic infection. It has been demonstrated as an effective alternative to surgical evacuation for the management of a confirmed first trimester miscarriage.

Contra-indications to Misoprostol:

- Severe asthma, hypertension, glaucoma, sickle cell anaemia, mitral stenosis, allergy to Misoprostol

Women should be informed of following:

- Increased bleeding and pain may be experienced
- Bleeding may continue for up to 3 weeks
- Oral analgesia protocols
- Side effects (usually self-limiting and last a few hours):
 - Fever >37.5 in 50%
 - Nausea and vomiting in 25%
 - Diarrhoea in 5%

- Rare risk of uterine rupture (incidence increased if prior uterine surgery)
 - This risk (and potential need for hysterectomy) **must** be specifically discussed in the pre-procedure counselling for all women

Treatment may be offered as an outpatient or day procedure:

- Procedure consent form should be completed and discussion documented in medical notes for all patients
- Patient requires an FBC and current Group and Hold
- If outpatient procedure, initial dose given in ED prior to discharge
 - eReferral to the [Early Pregnancy Assessment Service \(EPAS\)](#) at Armadale Hospital for follow up and subsequent Misoprostol dosing (if required)
 - Provide with safety netting and [discharge information](#)
- Gynaecology team to organise day procedure and follow up if this method chosen

4.9.3 Misoprostol Protocol (outpatient regime for early pregnancy loss < 14 weeks)

Day 1 (initial dose)

- Misoprostol 800microg (4x 200microg)
- Administer: PV (high in posterior fornix – use only water as lubricant) or PO or SUBLING

Day 7 or 14 (if required)

- Misoprostol 800microg (4x 200microg)
- Administer: PV (high in posterior fornix - use only water as lubricant) or PO or SUBLING

4.9.4 Surgical Management

Surgical management is indicated for women with a non-viable IUP if:

- The woman is haemodynamically unstable
- Has persistent excessive bleeding
- There is evidence of infected retained tissue
- Suspected Gestational Trophoblastic Disease
- Unsuccessful medical or expectant management
- The woman expresses a preference for the surgical option

The Gynaecology team will arrange admission and follow up if decision made for surgical management. If after hours, or urgent OT required, ED to contact the nurse manager.

4.10 Rh D Immunoglobulin for Sensitising Event Immunoprophylaxis

Confirm whether patient is Rhesus negative. A current (in last 2 weeks) Group and Screen must be available for the patient. A Kleihauer test is not required up to 20 weeks of pregnancy.

- Refer to [Rhesus negative and clinically significant antibodies in pregnant women AKG-PRO 0399/22](#)

The National Blood Authority/RANZCOG Guidelines (2021): [Prophylactic use of Rh D immunoglobulin in pregnancy care](#)^{14,15} lists the following sensitising events in the first 12 weeks of pregnancy:

- Miscarriage
 - “Insufficient evidence for routine use of Rh D immunoglobulin in threatened

miscarriage (ongoing pregnancy with uterine bleeding) unless bleeding repeated, heavy or associated with abdominal pain or significant pelvic trauma"¹⁴.

- Ectopic pregnancy
- Termination of pregnancy (medical > 10 weeks gestation, or surgical)
- Molar pregnancy
- Chorionic villus sampling

Rh D Immunoglobulin should be given as soon as practical within 72 hours of sensitising event¹⁴

- If delayed beyond 72 hours should be given up to 10 days from sensitising event, but may have lower efficacy

Dose in first 12 weeks of pregnancy^{14,15}:

- 250 IU given via slow deep intramuscular injection
- If gestational age is uncertain (possibly ≥ 13 weeks gestation) or multiple pregnancy then dose should be increased to 625 IU
- If ongoing uterine bleeding a repeat dose may be appropriate after an interval of 6 weeks (250 IU if during first 12 weeks gestation, 625 IU if ≥ 13 weeks gestation)

Consent:

- Informed consent should be obtained from the patient or legal guardian by the treating clinician for the administration of blood and blood products

4.11 Early Pregnancy Assessment Service (EPAS)

Currently running weekly on Tuesday mornings/Thursday afternoons (with an occasional Monday session each month)

Who to refer:

- Gestation <13+6/40 only (EPAS cannot see gestation 14 weeks and over)
- All miscarriages, including complete miscarriage (unless β -hCG ≤ 5)
- Pregnant women with PV bleeding in 1st trimester

Discuss with Gynaecology team prior to referral for:

- Late 1st trimester missed/incomplete miscarriages - may be more appropriate for surgical management

Referral to EPAS clinic at Armadale can be made via eReferral (referrals are triaged 7 days a week) and should include the following information as a minimum:

- Gravida/Parity and current gestation (indicate whether this is by LMP or USS)
- Clinical issue – brief summary of presentation
- Investigation results:
 - Serum β -hCG (plus previous result and date if available)
 - Blood group – if negative indicate whether RhD immunoglobulin has been given
 - Summary of any USS reports

All women referred to EPAS should be given a pathology form to have a repeat β -hCG in 48-72 hours i.e. a repeat result available prior to EPAS appointment

4.12 Early Pregnancy Loss Support Resources

Any woman being informed about potential/actual early pregnancy loss should be given the opportunity to have a support person (e.g., partner, relative, friend) with them at the time of receiving the news. This may necessitate an **exemption to the “no visitor policy”** in certain clinical areas of ED

Early Pregnancy Loss information packs are available in ED – these should only be given to women with CONFIRMED pregnancy loss. If packs are unavailable then copies of the resources included can be printed from links below:

- [Pregnancy Loss \(RANZCOG/SANDS\)](#)
- [Pink Elephants "Sorry for your loss" Miscarriage Care Kit](#)
- [Bereavement Services Brochure \(RedNose/SANDS\)](#)

Other recommended resources for patients:

- Sands www.sands.org.au
 - Miscarriage, stillbirth and neonatal death support for parents and healthcare professionals
 - Fact sheets and multicultural resources available online
 - 24/7 Bereavement Support Line: 1300 308 307
- Red Nose <https://rednose.org.au/>
 - Bereavement support services and resource library
- The Pink Elephants Support Network www.pinkelephants.org.au
 - Not-for-profit organisation (ACNC Registered Charity) providing resources, information and peer-support for anyone impacted by early pregnancy loss
- Pregnancy, Birth and Baby www.pregnancybirthbaby.org.au/pregnancy-loss
 - Government funded website operated by HealthDirect

Recognition Certificates for early pregnancy loss are available from the Western Australian Registry of Births Deaths and Marriages for babies that are not able to be formally registered under the Births, Deaths and Marriages Registration Act 1998 i.e. loss took place before 20 weeks gestation. The certificates are free.

- [Application form](#)
- Print out and give to parent(s) if they would like to apply for recognition
- Treating medical practitioner or midwife must sign the Health Professional's declaration on page 4

5. Legislative context

The following documents are required to give effect to this policy:

- Health Department of Western Australia (HDWA) [Consent to Treatment OD 0657/16](#)
- HDWA [Recognising and Responding to Acute Deterioration Policy MP 0086/18](#)

6. Supporting information

The following documents support the implementation of this policy:

- [Medication Management Procedure AKG-PRO-0255-18](#)
- [AKG Massive Transfusion Protocol: Obstetric](#)
- [Rhesus negative and clinically significant antibodies in pregnant women AKG-PRO-0399/22](#)

7. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition – updated May 2021):

- Standard 2 – Action 2.03
- Standard 8 – Action 8.04
- Standard 8 – Action 8.06

8. References

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13. [KEMH Clinical Practice Guidelines, Gynaecology: Restricted Area Guideline: Miscarriage \(2019\)](#)
14. National Blood Authority (NBA) guideline 2021 [Guideline for the prophylactic use of Rh D immunoglobulin in pregnancy care.pdf \(blood.gov.au\)](#)
15. RANZCOG Guideline 2021 [Guidelines for the use of Rh\(D\) Immunoglobulin \(Anti-D\) in obstetrics \(ranzcoг.edu.au\)](#)

9. Document contact

Consultant Emergency Physician

Enquiries relating to this guideline may be directed to:

Title: Dr Emma Trace, Consultant Emergency Physician
 Department: Emergency Department
 Email: emma.trace@health.wa.gov.au

10. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

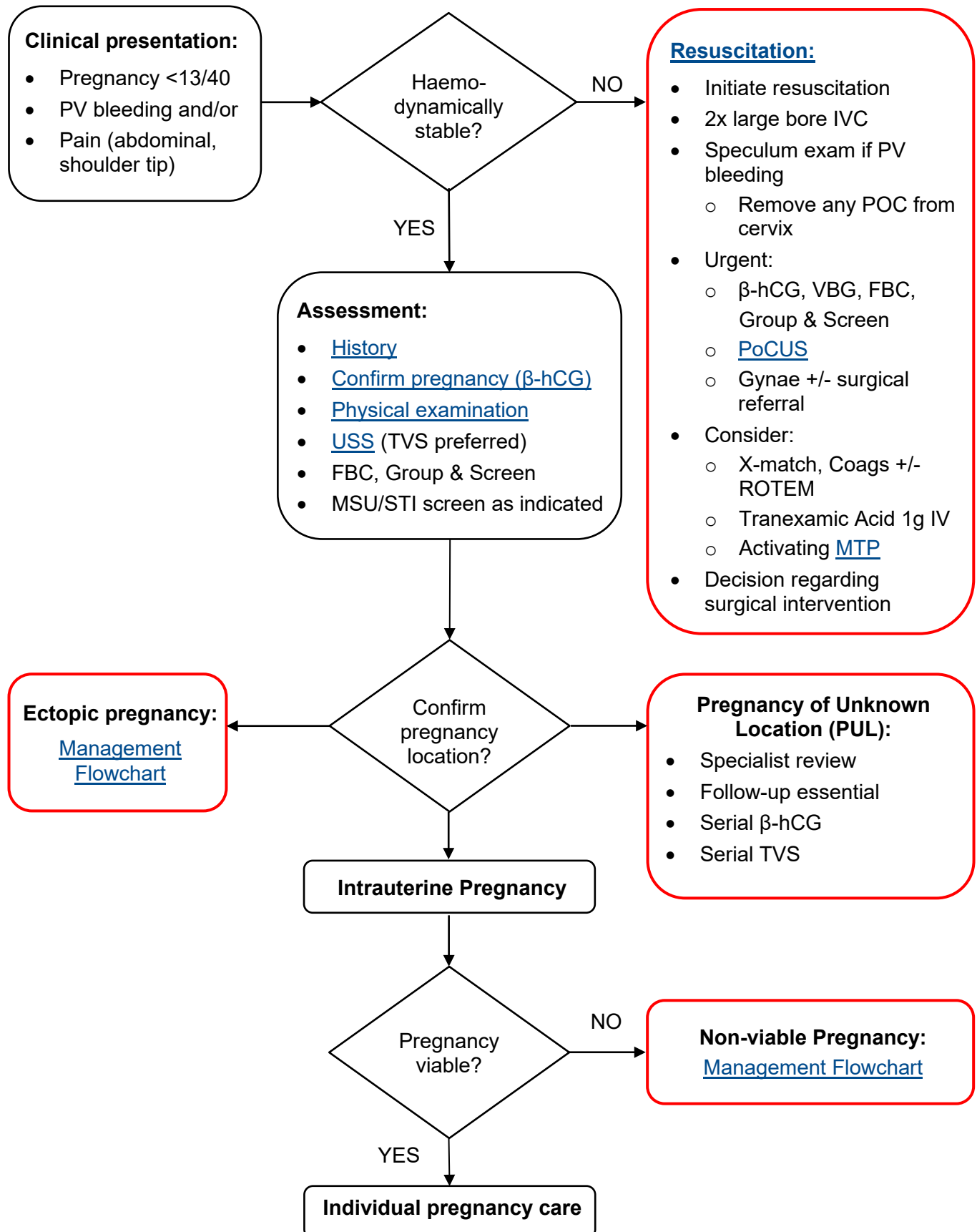
Version	Effective from	Effective to	Amendment(s)
V2	23/05/2023	23/05/2026	Scheduled review

11. Authorisation

This guideline has been authorised and issued by the Director Clinical Services.

Authorised by	Dr Xin Chau – A/Director Clinical Services
Authorisation date	23/05/2023
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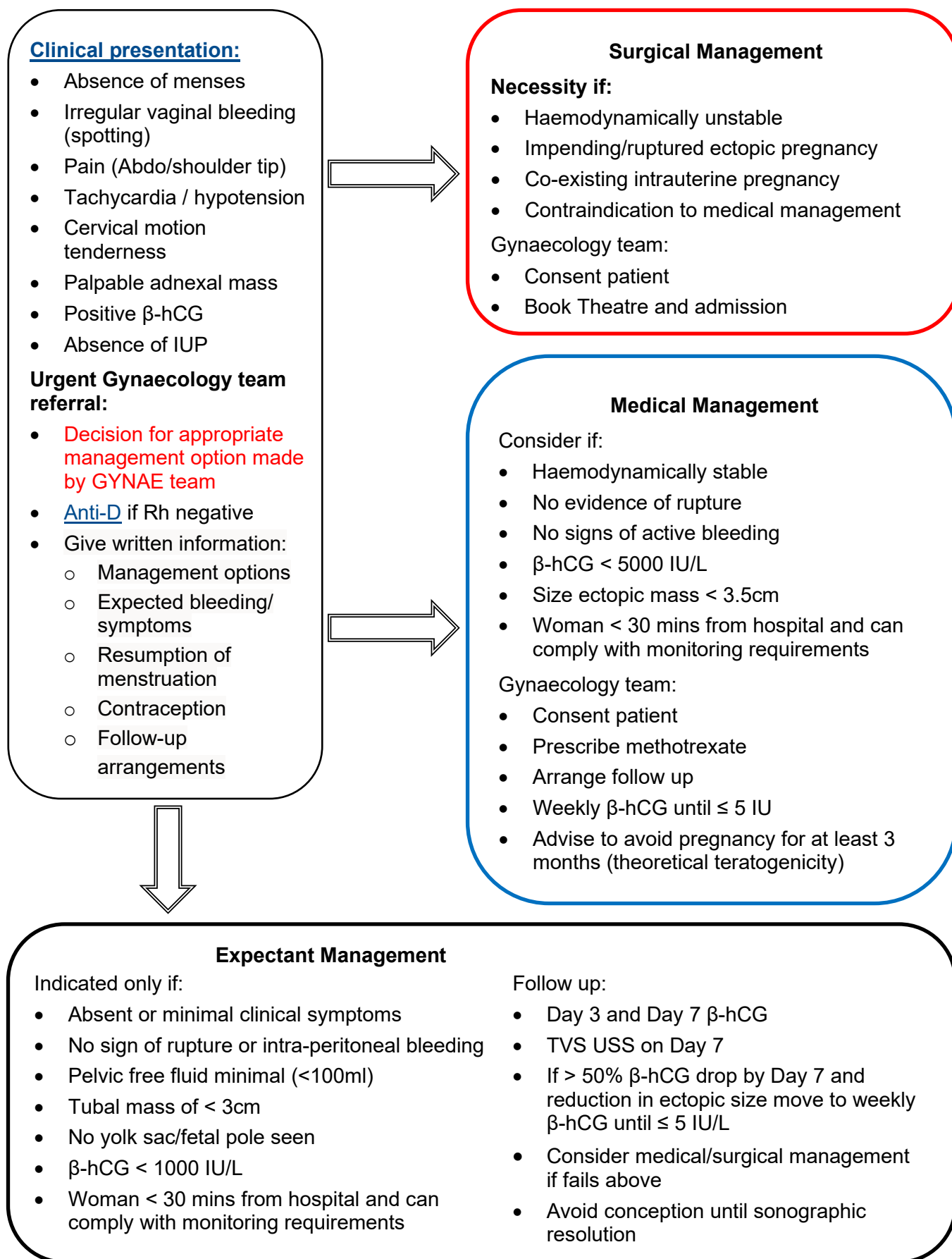
Appendix 1 - Assessment of Suspected Early Pregnancy Loss Flow Chart



Appendix 2 - Correlation of Ultrasound with Serum β -hCG Measurements

B-hCG	TVS	Recommendation
≤ 1500 IU/L	Expect empty uterus Look for free fluid and adnexal mass	Free fluid +/- adnexal mass present: - high probability of ectopic pregnancy Free fluid +/- adnexal mass absent: - repeat β-hCG in 48 hours - TVS when β-hCG reaches/expected to reach discriminatory zone
> 1500 IU/L	No IUP seen with adnexal mass or free fluid	High probability of ectopic pregnancy refer to Ectopic Pregnancy
> 1500 IU/L	No IUP seen with no other abnormal findings	Still high probability of ectopic but probably stable. Might be spontaneous tubal abortion or spontaneous failing pregnancy repeat β-hCG +/- TVS in 48 hours
Declining or sub-optimally rising β -hCG levels (as per above thresholds)	No intrauterine pregnancy seen	Indicates a non-viable pregnancy (IUP or ectopic) ensure appropriate follow up with weekly β-hCG until ≤ 5 IU/L

Appendix 3: Management of Ectopic Pregnancy Flow Chart



Appendix 4: Management of Non-Viable Intrauterine Pregnancy Flow Chart

